

Deep Dive: 6. Loops

Concept Summary

Loops allow you to repeat a block of code multiple times, either for a set number of times (for) or until a condition changes (while).

Coding Structure

```
for item in sequence:
    # code

while condition:
    # code
```

Example Code

```
for i in range(3):
    print(f'Count: {i}')

battery = 10
while battery > 0:
    battery -= 5
```

Code Explanation

- Line 1: 'range(3)' generates numbers 0, 1, 2. The loop runs 3 times.
- Line 2: Prints the current value of i.
- Line 5: A while loop that runs as long as battery is greater than 0.
- Line 6: Subtracts 5 from battery each loop. It runs twice before stopping.

Practice Problems

1. Write a for loop that prints all even numbers from 2 to 10.
2. Write a while loop that simulates a countdown timer from 5 to 0, printing 'Blastoff!' at 0.